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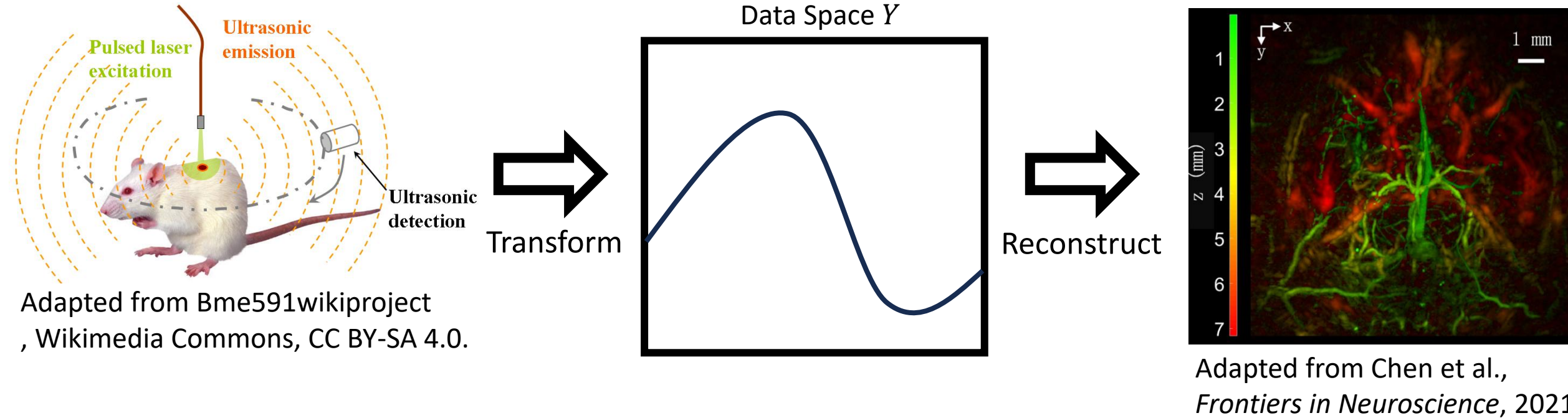
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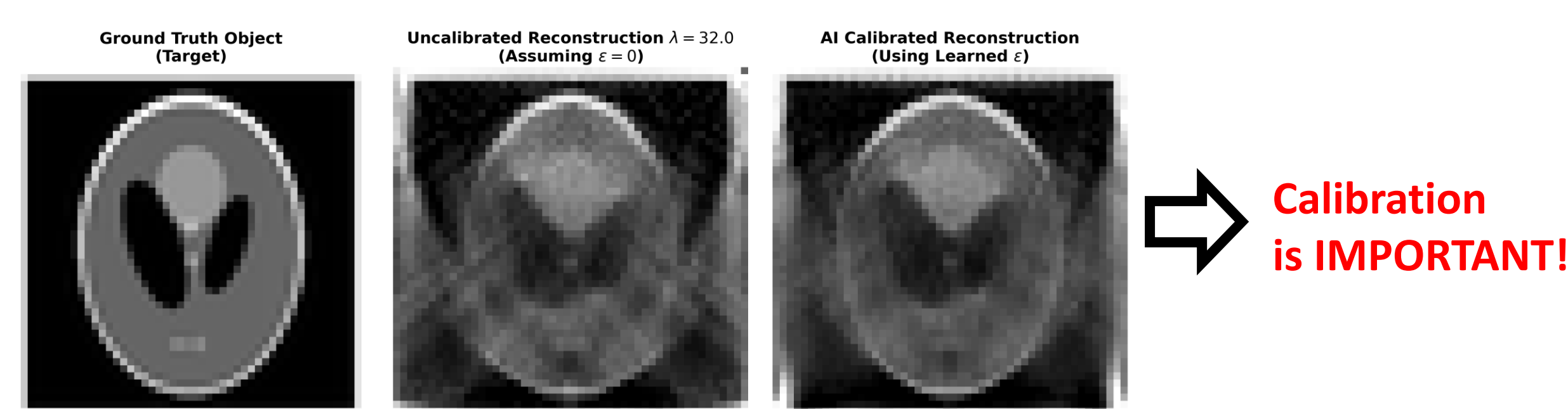
Tomographic Imaging

- What is Tomography?
 - Slice imaging technique that focus on **integration**
 - X-ray: Line; Ultrasound/PAT: Circle

- What is **Photoacoustic Tomography**?



- Why this problem?
 - Standard image reconstruction assumes perfectly known sensor locations.
 - Unknown sensor displacements** in reality severely degrade image quality.



- Why not manually?
 - Time consuming
 - Varies by machine
 - Requires experience

Learning-based method $\Rightarrow$ **Higher accuracy** • **Automated** • **Robust**

Levelset Representation

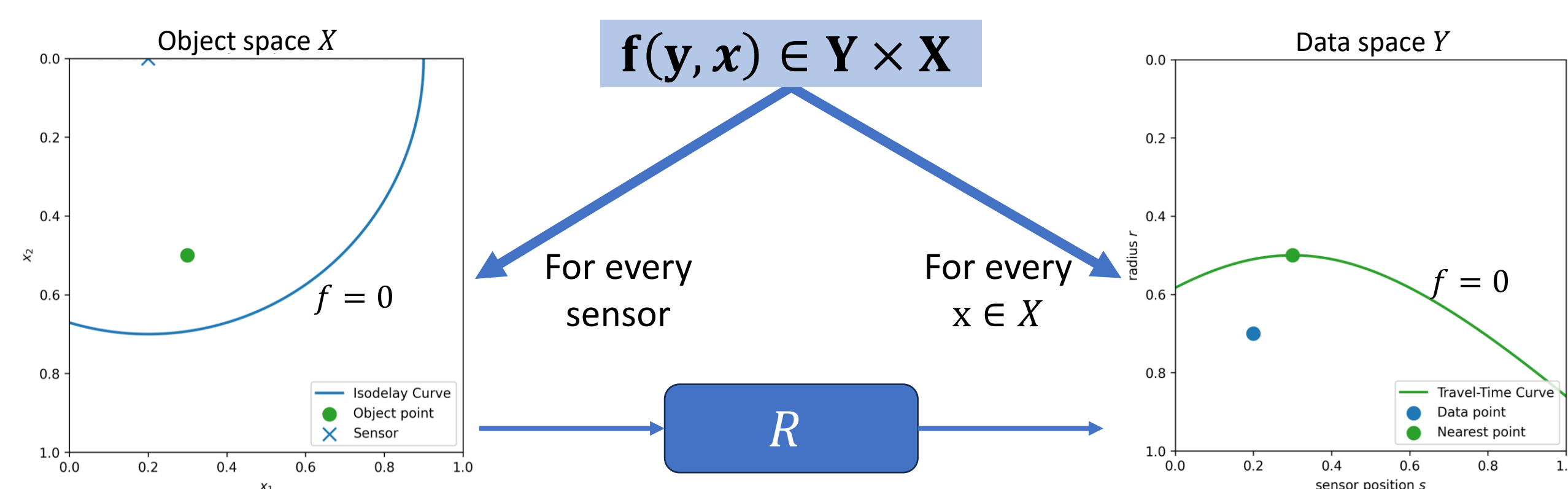
Tomography Mathematics

$Ru = \int_X u(x) \delta(f(y, x)) dx$ $f(y, x)$: geometry to integrate along

For X-ray: $f(y, x) = \langle \vec{\theta}_y, x \rangle - S_y$

For Ultrasound: $f(y, x) = \sqrt{(x_1 - x_s)^2 + x_2^2} - r$

$u(x)$: image



Photoacoustic Tomography

$$Ru(x', t) = \partial_t \left[\frac{1}{t} \int_X u(x) \delta(t - |x - x'|) dx \right] = v$$

Gaussian Approx. $\delta \rightarrow Ru_\lambda \approx w(y) \frac{\int_X e^{-\lambda |f(y, x)|^2} u(x) dx}{\int_X e^{-\lambda |f(y, x)|^2} dx}$

Image Reconstruction:

$$u = (R^*R + \alpha I)^{-1} R^*v$$

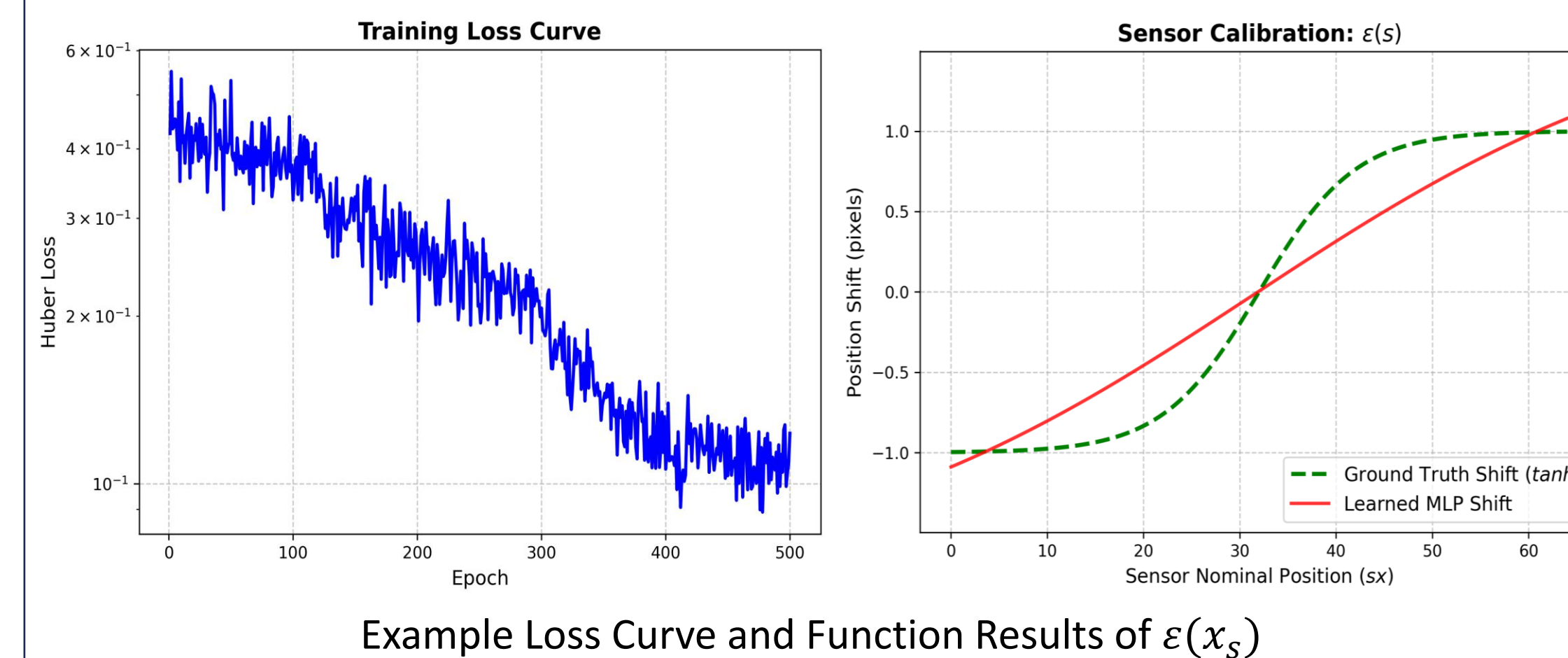
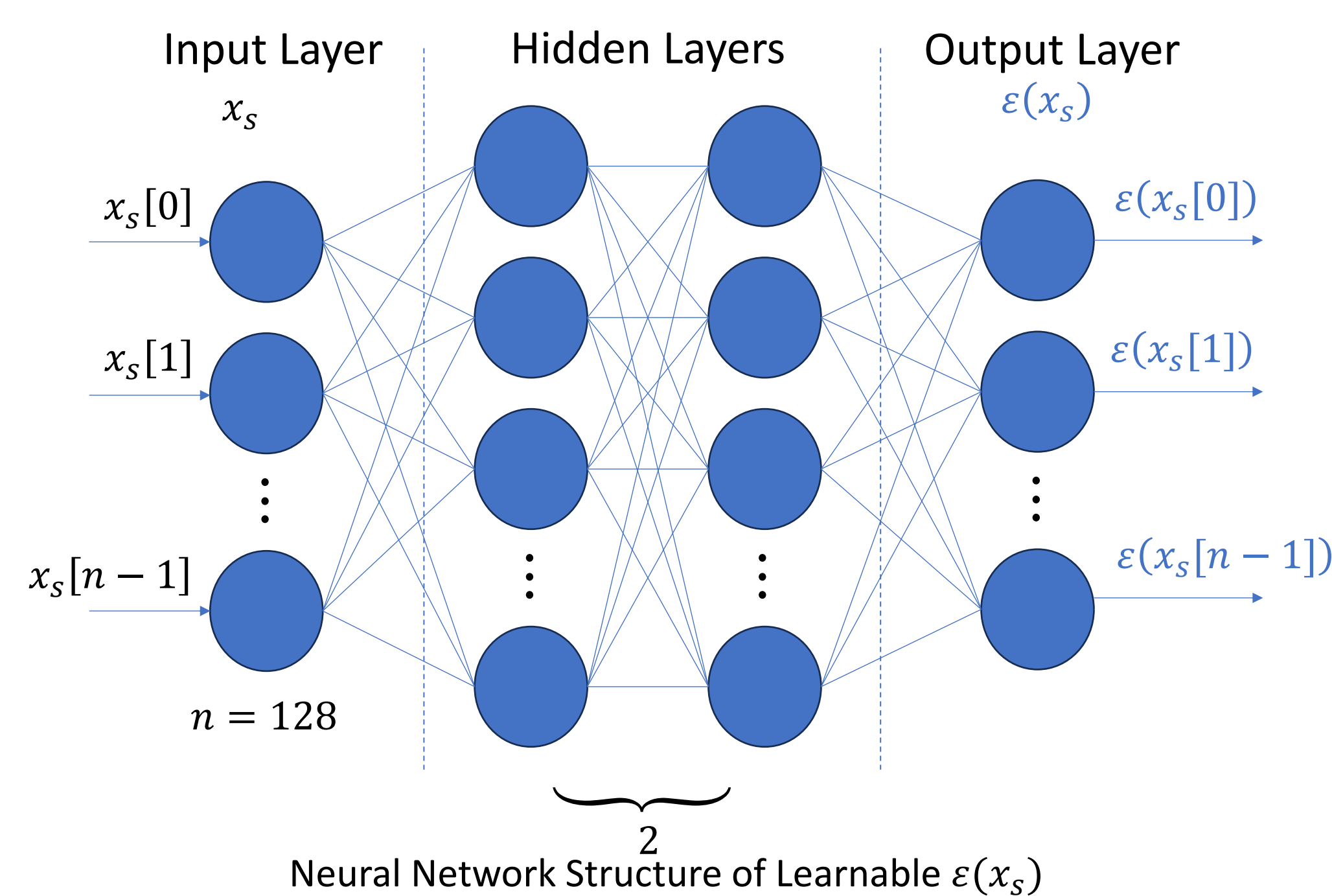
- R^*R : Simulate blurring of physical system
- R^*v : Project $Y \rightarrow X$
- αI : Regularizer $\rightarrow$ numerical stability

Learning and Calibration

- Problem:**
PAT machines with displaced sensors $\rightarrow$ artifacts & blurring
 $\Rightarrow$ **Let f be LEARNABLE!**

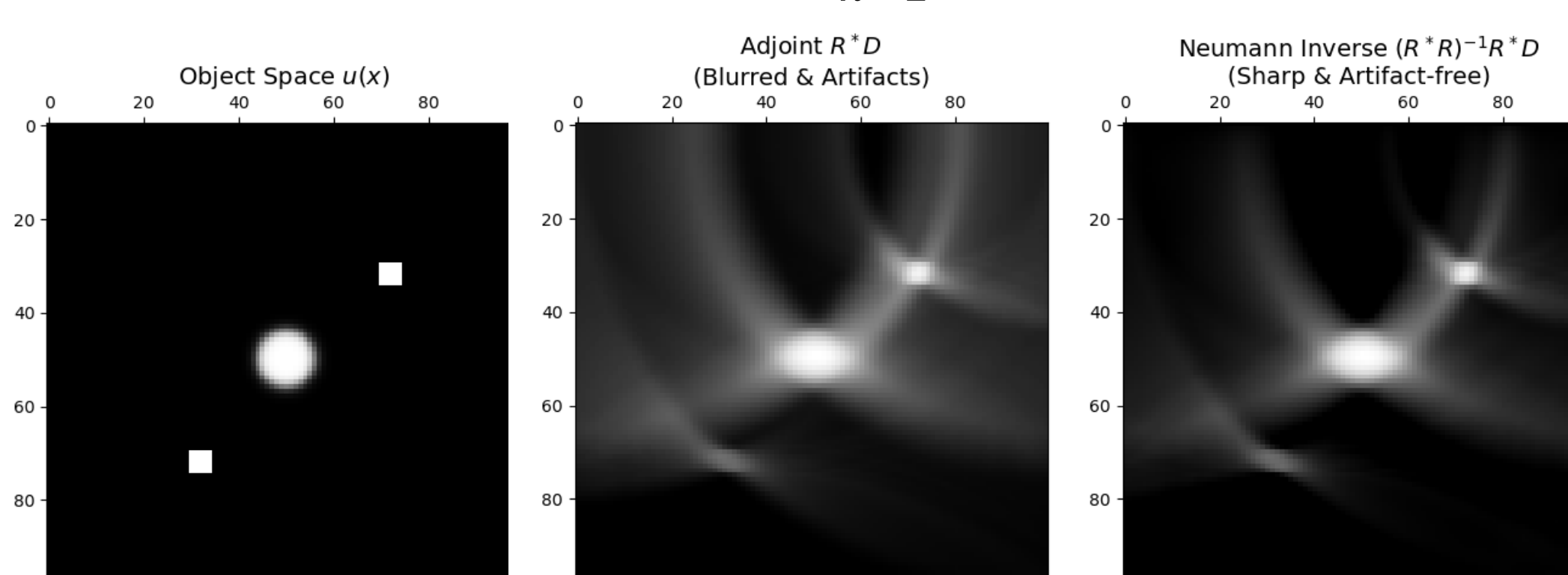
Assume solid physics
 $\rightarrow$ perturbation is only related to sensor location jitter.
 $\rightarrow \tilde{f}(y, x) = f(x_s + \epsilon(x_s), r; x)$

Learnable Perturbation $\epsilon(x_s)$



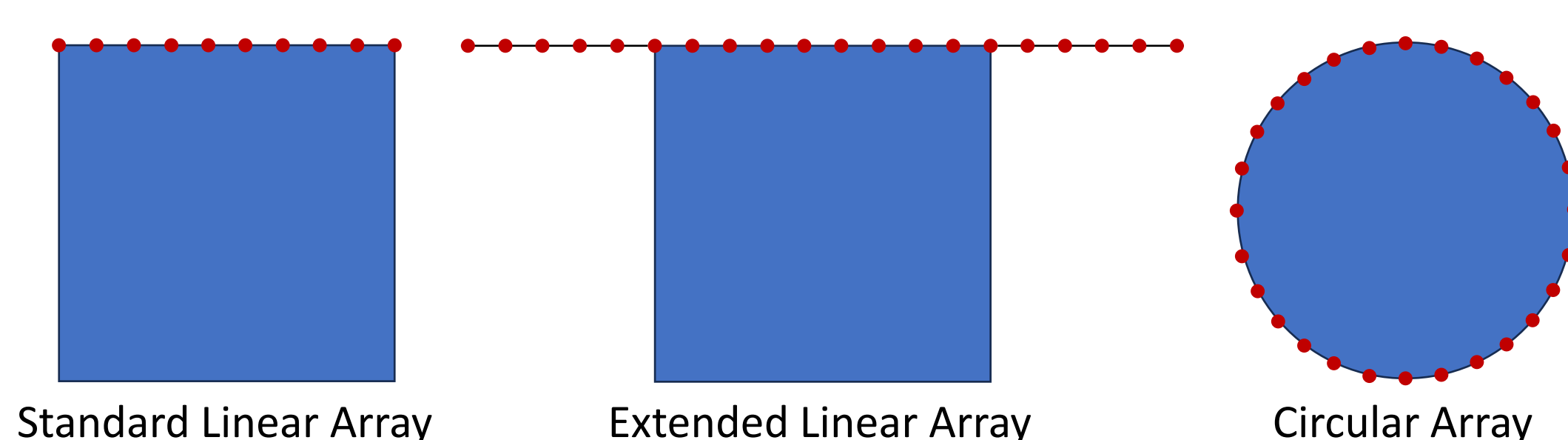
Neumann Series: $(I - A)^{-1} = I + A + A^2 + A^3 + \dots$

$$(R^*R + \alpha I)^{-1} = I + \sum_{k=1}^{T-1} (I - R^*R - \alpha I)^k$$



Sensor Alignments

- Why?**
 - Standard linear arrays: **limited-view**
 - Fail to reconstruct certain features.

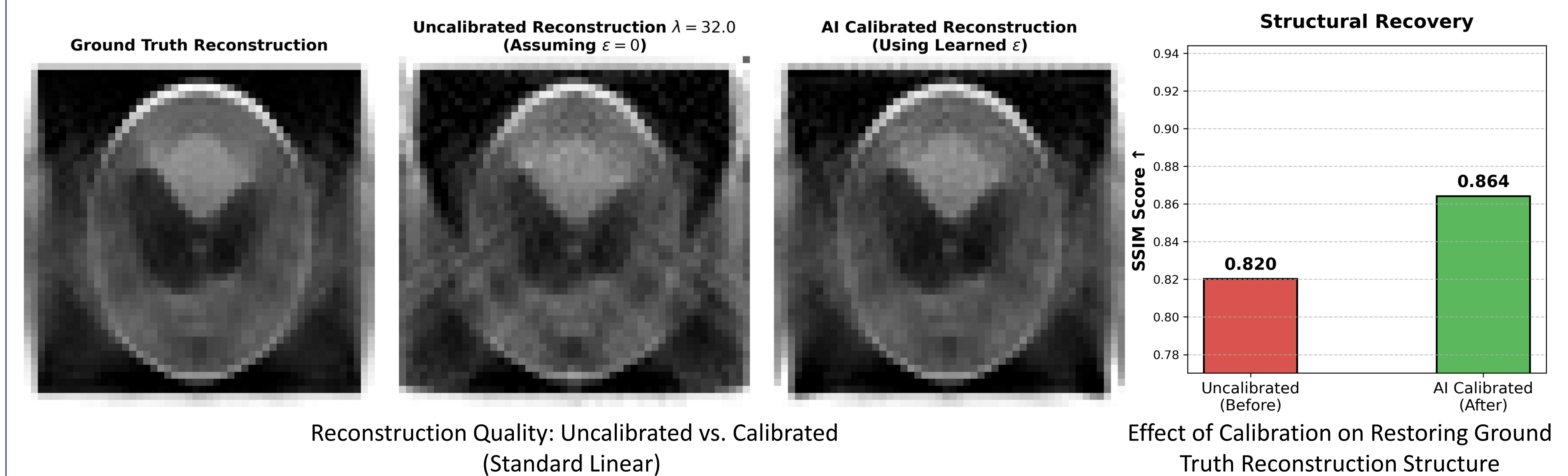


Results

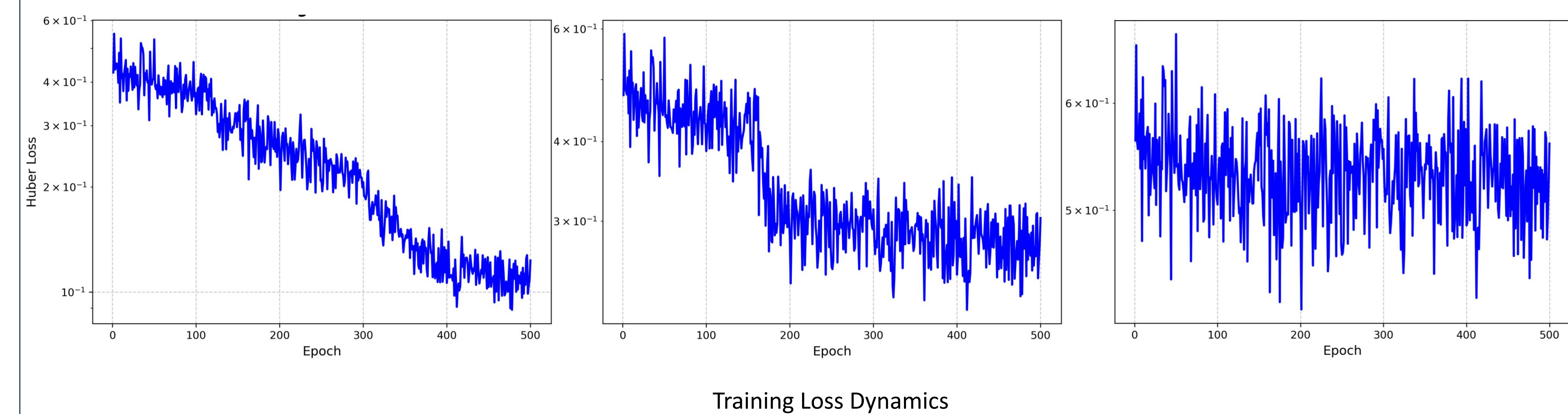
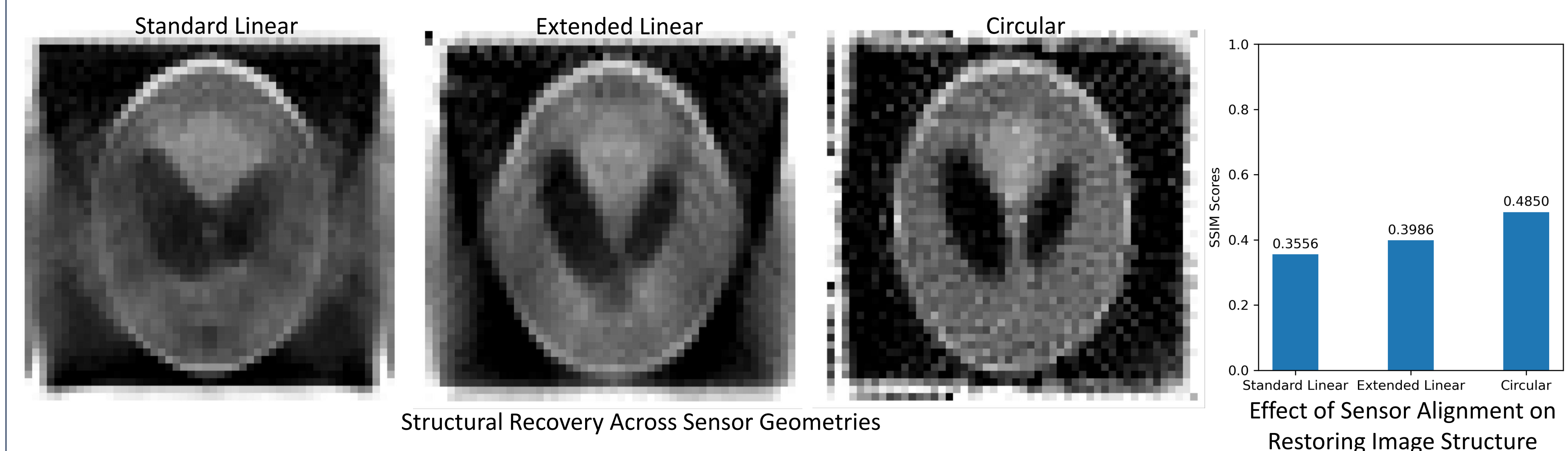
Evaluation Criterion

- Structural Similarity Measure (SSIM): Compares structural information, not pixel difference
- Huber Loss:
 - Similar to Mean Squared Error for small errors
 - Linear for large

Effect of Calibration



Effect of Sensor Alignment



Conclusions & Future Works

Conclusions

- Learning-based calibration successfully eliminates artifacts.
- Extended linear arrays capture finer details, circular arrays trace edges better.
- Circular arrays achieve the highest SSIM despite higher training losses.

Future Works

- Involve actual PAT sensor data to improve training quality.
- Try combining algorithms.
- Tune the hyperparameters more to optimize model performance.